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e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – B)** | **Time:** 1 Hour

Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 30

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: 1 Mark Questions (3 MCQs + 2 A–R) MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The distance between the points $(m, -n)$ and $(-m, n)$ is: [1]

(A) $\sqrt{m^2+n^2}$ **(B)** $m+n$ **(C)** $2\sqrt{m^2+n^2}$ **(D)** $\sqrt{2m^2+2n^2}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) $2\sqrt{m^2+n^2}$

Given: Points $A(m, -n)$ and $B(-m, n)$.

We know that the distance between two points (x_1, y_1) and (x_2, y_2) is:

Formula Used: $AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Substituting $x_1 = m$, $y_1 = -n$, $x_2 = -m$, and $y_2 = n$:

$$AB = \sqrt{(-m-m)^2+(n-(-n))^2}$$

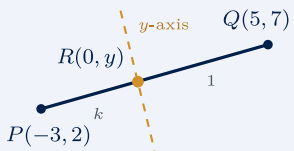
$$= \sqrt{(-2m)^2+(2n)^2} = \sqrt{4m^2+4n^2} = \sqrt{4(m^2+n^2)} = \boxed{\mathbf{2\sqrt{m^2+n^2}}}$$

Q2. The line segment joining the points $P(-3, 2)$ and $Q(5, 7)$ is divided by the y-axis in the ratio: [1]

(A) 3 : 5 **(B)** 5 : 3 **(C)** 2 : 7 **(D)** 7 : 2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 3 : 5



Given: $P(-3, 2)$ and $Q(5, 7)$. Let the y-axis intersect segment PQ at $R(0, y)$ in the internal ratio $k : 1$.

We know the internal section formula:

Formula Used: $x = \dfrac{m_1x_2 + m_2x_1}{m_1+m_2}$

Since the x-coordinate of any point on the y-axis is 0:

$$\frac{k(5)+1(-3)}{k+1} = 0 \implies 5k-3=0 \implies 5k=3 \implies k=\frac{3}{5}$$

Required ratio $= \mathbf{3 : 5}$

Q3. The coordinates of the midpoint of the line segment joining the points $(-5, 4)$ and $(7, -8)$ are: [1]

(A) $(1, -2)$ **(B)** $(-1, 2)$ **(C)** $(2, -4)$ **(D)** $(6, -6)$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** $(1, -2)$

Given: Points $(-5, 4)$ and $(7, -8)$.

We know the midpoint formula for (x_1, y_1) and (x_2, y_2):

Formula Used: $M(x, y) = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$

Substituting the coordinates:

$$M = \left(\frac{-5 + 7}{2}, \frac{4 + (-8)}{2}\right) = \left(\frac{2}{2}, \frac{-4}{2}\right) = \boxed{(\mathbf{1, -2})}$$

Q4. **Assertion (A):** The distance of the point $(0, 8)$ from the origin is 8 units.
Reason (R): The distance of any point (x, y) from the origin is given by $\sqrt{x^2 + y^2}$. [1]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)**

Evaluating Reason (R): The distance between the origin $(0, 0)$ and any point (x, y) is:

Formula Used: $d = \sqrt{(x - 0)^2 + (y - 0)^2} = \sqrt{x^2 + y^2}$

Thus, Reason (R) is **true**.

Evaluating Assertion (A): Substituting $(x, y) = (0, 8)$ into the formula stated in Reason (R):

$$d = \sqrt{0^2 + 8^2} = \sqrt{64} = \boxed{8 \text{ units}}$$

Thus, Assertion (A) is **true**, Reason (R) is **true**, and (R) is the correct explanation of (A).

Q5. **Assertion (A):** If the points $(1, 2)$, $(4, y)$, $(x, 6)$ and $(3, 5)$ are the vertices of a parallelogram taken in order, then $x = 6$ and $y = 3$.
Reason (R): The diagonals of a parallelogram bisect each other. [1]

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Evaluating Reason (R): It is a fundamental geometric theorem that the diagonals of any parallelogram bisect each other. So Reason (R) is **true**.

Evaluating Assertion (A): Let $A(1, 2)$, $B(4, y)$, $C(x, 6)$, and $D(3, 5)$ be the vertices in order. Using Reason (R), midpoint of diagonal AC = midpoint of diagonal BD:

Formula Used: $\left(\frac{x_A + x_C}{2}, \frac{y_A + y_C}{2}\right) = \left(\frac{x_B + x_D}{2}, \frac{y_B + y_D}{2}\right)$

Substituting the coordinates:

$$\left(\frac{1 + x}{2}, \frac{2 + 6}{2}\right) = \left(\frac{4 + 3}{2}, \frac{y + 5}{2}\right)$$

Equating abscissae: $\frac{1 + x}{2} = \frac{7}{2} \implies 1 + x = 7 \implies \boxed{x = 6}$.

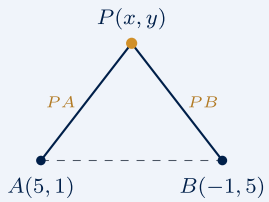
Equating ordinates: $\frac{8}{2} = \frac{y + 5}{2} \implies y + 5 = 8 \implies \boxed{y = 3}$.

Thus, Assertion (A) is **true** and Reason (R) is the correct explanation of (A).

SECTION B: Short Answer Type Questions — MATHS MANTHAN | (5 × 2 = 10 Marks)

Q6. If the point $P(x, y)$ is equidistant from the points $A(5, 1)$ and $B(-1, 5)$, prove that $3x = 2y$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Distance Equality Condition [0.5 Mark]
Given: $PA = PB \implies PA^2 = PB^2$ for $P(x, y)$, $A(5, 1)$, and $B(-1, 5)$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Algebraic Expansion and Proof [1.5 Marks]
Using the distance formula for $PA^2 = PB^2$:

$$(x - 5)^2 + (y - 1)^2 = (x - (-1))^2 + (y - 5)^2$$

Expanding both sides with $(a \pm b)^2 = a^2 \pm 2ab + b^2$:

$$x^2 - 10x + 25 + y^2 - 2y + 1 = x^2 + 2x + 1 + y^2 - 10y + 25$$

Cancelling x^2, y^2, 25, and 1 from both sides:

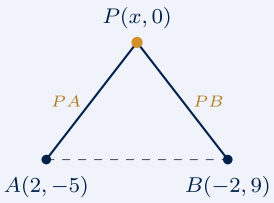
$$-10x - 2y = 2x - 10y \implies -10x - 2x = -10y + 2y \implies -12x = -8y$$

Dividing both sides by -4:

$$\boxed{\mathbf{3x = 2y} \quad \text{(Hence Proved)}}$$

Q7. Find the coordinates of the point on the x-axis which is equidistant from the points $(2, -5)$ and $(-2, 9)$. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Coordinate Representation & Condition [0.5 Mark]

Given: $A(2, -5)$, $B(-2, 9)$. Any point on the x-axis has the form $P(x, 0)$. Since $PA = PB$, we have $PA^2 = PB^2$.

We know the squared distance formula:

Formula Used: $d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Step 2: Solving for x [1.5 Marks]

Substituting $P(x, 0)$, $A(2, -5)$, and $B(-2, 9)$ into $PA^2 = PB^2$:

$$(x - 2)^2 + (0 - (-5))^2 = (x - (-2))^2 + (0 - 9)^2$$

$$(x^2 - 4x + 4) + 25 = (x^2 + 4x + 4) + 81$$

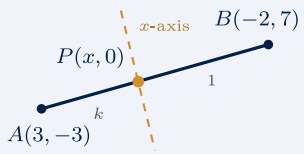
Cancelling $x^2 + 4$ from both sides:

$$-4x + 25 = 4x + 81 \implies -4x - 4x = 81 - 25 \implies -8x = 56 \implies x = \frac{56}{-8} = -7$$

Required point on the x-axis $= (\mathbf{-7}, \mathbf{0})$

Q8. Find the ratio in which the line segment joining the points $A(3, -3)$ and $B(-2, 7)$ is divided by the x-axis. Also find the coordinates of the point of division. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Finding Ratio from y-Coordinate [1 Mark]

Given: $A(3, -3)$, $B(-2, 7)$. A point on the x-axis is $P(x, 0)$. Let P divide AB in the ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

Equating the y-coordinate to 0:

$$\frac{k(7) + 1(-3)}{k + 1} = 0 \implies 7k - 3 = 0 \implies 7k = 3 \implies k = \frac{3}{7} \implies \text{Ratio} = \mathbf{3 : 7}$$

Step 2: Finding Coordinates of P [1 Mark]

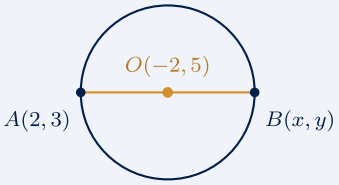
Substituting $k = \frac{3}{7}$ (or $m_1 = 3$, $m_2 = 7$), $x_1 = 3$, and $x_2 = -2$ into the x-coordinate formula:

$$x = \frac{3(-2) + 7(3)}{3 + 7} = \frac{-6 + 21}{10} = \frac{15}{10} = \frac{3}{2}$$

Ratio $= \mathbf{3 : 7}$ and Point of division $= \left(\mathbf{\frac{3}{2}}, \mathbf{0} \right)$

Q9. If the coordinates of one end of a diameter of a circle are $(2, 3)$ and the coordinates of its centre are $(-2, 5)$, find the coordinates of the other end of the diameter. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Midpoint Relation [0.5 Mark]
Given: Endpoint $A(2, 3)$ and centre $O(-2, 5)$. Let the other endpoint be $B(x, y)$. Centre O is the midpoint of diameter AB.

We apply the midpoint formula:

Formula Used: $O = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Step 2: Solving for x and y [1.5 Marks]
Equating centre $O(-2, 5)$ with the midpoint of AB:

$$\left(\frac{2 + x}{2}, \frac{3 + y}{2}\right) = (-2, 5)$$

Comparing both coordinates:

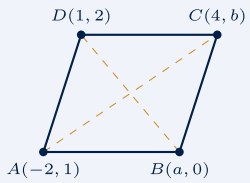
$$\frac{2 + x}{2} = -2 \implies 2 + x = -4 \implies x = -4 - 2 = -6$$

$$\frac{3 + y}{2} = 5 \implies 3 + y = 10 \implies y = 10 - 3 = 7$$

Other end of the diameter $= (\mathbf{-6}, \mathbf{7})$

Q10. If $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$, find the values of a and b. Hence, find the length of side AB. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Finding a and b Using Diagonals [1 Mark]
Given: Parallelogram $ABCD$ with $A(-2, 1)$, $B(a, 0)$, $C(4, b)$, $D(1, 2)$. Diagonals AC and BD bisect each other, so Midpoint of AC = Midpoint of BD.

We use the midpoint and distance formulas:

Formula Used: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ and $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Equating the midpoints of diagonals AC and BD:

$$\left(\frac{-2 + 4}{2}, \frac{1 + b}{2}\right) = \left(\frac{a + 1}{2}, \frac{0 + 2}{2}\right) \implies \left(1, \frac{1 + b}{2}\right) = \left(\frac{a + 1}{2}, 1\right)$$

Comparing coordinates: $\frac{a + 1}{2} = 1 \implies a + 1 = 2 \implies \mathbf{a = 1}$, and $\frac{1 + b}{2} = 1 \implies 1 + b = 2 \implies \mathbf{b = 1}$.

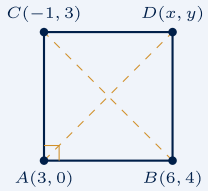
Step 2: Finding Length of Side AB [1 Mark]
With $a = 1$, B is $(1, 0)$. Distance from $A(-2, 1)$ to $B(1, 0)$ is:

$$AB = \sqrt{(1 - (-2))^2 + (0 - 1)^2} = \sqrt{3^2 + (-1)^2} = \sqrt{9 + 1} = \sqrt{10} \text{ units}$$

$\mathbf{a = 1}$, $\mathbf{b = 1}$ and $\mathbf{AB = \sqrt{10}}$ units

Q11. The vertices of a triangle ABC are $A(3,0)$, $B(6,4)$ and $C(-1,3)$.
(i) Calculate the lengths of sides AB, BC and AC, and prove that $\triangle ABC$ is a right-angled isosceles triangle. [3]
(ii) Find the coordinates of the fourth vertex D such that $ABCD$ forms a square with diagonal BC. [2] [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



(i) Side Lengths and Right-Angled Isosceles Proof [3 Marks]
Given: $A(3,0)$, $B(6,4)$, $C(-1,3)$. We use the distance formula to compute all three sides and verify Pythagoras' theorem.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$$AB = \sqrt{(6-3)^2+(4-0)^2} = \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = \mathbf{5} \text{ units}$$
$$AC = \sqrt{(-1-3)^2+(3-0)^2} = \sqrt{(-4)^2+3^2} = \sqrt{16+9} = \sqrt{25} = \mathbf{5} \text{ units}$$
$$BC = \sqrt{(-1-6)^2+(3-4)^2} = \sqrt{(-7)^2+(-1)^2} = \sqrt{49+1} = \sqrt{50} = \mathbf{5\sqrt{2}} \text{ units}$$

Since $AB = AC = 5$ units, $\triangle ABC$ is **isosceles**. Moreover, checking Pythagoras' condition:

$$AB^2 + AC^2 = 5^2 + 5^2 = 25 + 25 = 50 = (\sqrt{50})^2 = BC^2$$

By the converse of Pythagoras' theorem, $\angle A = 90°$. Hence $\triangle ABC$ is a **right-angled isosceles triangle**.

(ii) Finding Fourth Vertex D for Square with Diagonal BC [2 Marks]
Because BC is a diagonal of the square, the opposite pairs of vertices are B, C and A, D. Since diagonals of a square bisect each other:

Formula Used: Midpoint of AD = Midpoint of BC
$$\left(\frac{x_A+x_D}{2}, \frac{y_A+y_D}{2}\right) = \left(\frac{x_B+x_C}{2}, \frac{y_B+y_C}{2}\right)$$

Let D be (x, y). Equating the midpoints of AD and BC:

$$\left(\frac{3+x}{2}, \frac{0+y}{2}\right) = \left(\frac{6+(-1)}{2}, \frac{4+3}{2}\right) = \left(\frac{5}{2}, \frac{7}{2}\right)$$

Comparing coordinates: $3 + x = 5 \implies x = 2$, and $y = 7$.

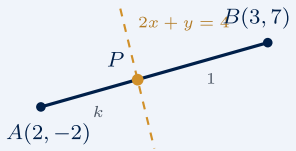
$$\boxed{\text{Sides: } AB = AC = 5,\ BC = 5\sqrt{2} \qquad \text{and} \qquad D = (\mathbf{2}, \mathbf{7})}$$

Q12. A straight line with equation $2x + y - 4 = 0$ intersects the line segment joining the points $A(2, -2)$ and $B(3, 7)$.

(i) Find the ratio in which the line divides the segment AB internally. [3]

(ii) Hence, determine the exact coordinates of the point of intersection. [2] [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



(i) Finding the Internal Ratio [3 Marks]

Given: $A(2, -2)$, $B(3, 7)$, and dividing line $2x + y - 4 = 0$. Let the line intersect AB at $P(x, y)$ in ratio $k : 1$.

We know the internal section formula:

Formula Used: $P(x, y) = \left(\dfrac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \dfrac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$

Using $m_1 = k$ and $m_2 = 1$, the coordinates of P in terms of k are:

$$P(x, y) = \left(\frac{3k + 2}{k + 1}, \frac{7k - 2}{k + 1} \right)$$

Since $P(x, y)$ lies on the line $2x + y - 4 = 0$, it must satisfy its equation:

$$2\left(\frac{3k + 2}{k + 1} \right) + \left(\frac{7k - 2}{k + 1} \right) - 4 = 0$$

Multiplying through by $(k + 1)$:

$$2(3k + 2) + (7k - 2) - 4(k + 1) = 0$$

$$6k + 4 + 7k - 2 - 4k - 4 = 0 \implies 9k - 2 = 0 \implies 9k = 2 \implies k = \frac{2}{9} \implies \text{Ratio} = \mathbf{2 : 9}$$

(ii) Coordinates of the Point of Intersection [2 Marks]

Substituting $k = \dfrac{2}{9}$ (or $m_1 = 2$, $m_2 = 9$) back into $P(x, y)$:

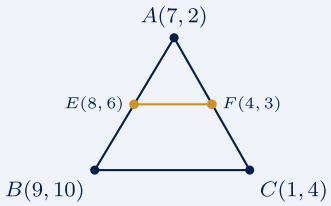
$$x = \frac{2(3) + 9(2)}{2 + 9} = \frac{6 + 18}{11} = \frac{24}{11}, \qquad y = \frac{2(7) + 9(-2)}{2 + 9} = \frac{14 - 18}{11} = -\frac{4}{11}$$

$$\boxed{\text{Ratio } = \mathbf{2 : 9} \qquad \text{and} \qquad P = \left(\mathbf{\frac{24}{11}}, \mathbf{-\frac{4}{11}} \right)}$$

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Q13. The coordinates of the vertices of $\triangle ABC$ are $A(7,2)$, $B(9,10)$ and $C(1,4)$. If E and F are the midpoints of sides AB and AC respectively:
(i) Find the coordinates of midpoints E and F. [2]
(ii) Calculate the lengths of segments EF and BC using the distance formula. [2]
(iii) Hence, verify and prove that $EF = \frac{1}{2}BC$. [1] [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



(i) Finding Midpoints E and F [2 Marks]
Given: $A(7,2)$, $B(9,10)$, $C(1,4)$. E is the midpoint of AB and F is the midpoint of AC.

We know the midpoint and distance formulas:

Formula Used: $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ and $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$$E = \left(\frac{7+9}{2}, \frac{2+10}{2}\right) = \left(\frac{16}{2}, \frac{12}{2}\right) = (\mathbf{8}, \mathbf{6})$$

$$F = \left(\frac{7+1}{2}, \frac{2+4}{2}\right) = \left(\frac{8}{2}, \frac{6}{2}\right) = (\mathbf{4}, \mathbf{3})$$

(ii) Lengths of EF and BC [2 Marks]
Using the distance formula for $E(8,6)$, $F(4,3)$ and $B(9,10)$, $C(1,4)$:

$$EF = \sqrt{(4-8)^2+(3-6)^2} = \sqrt{(-4)^2+(-3)^2} = \sqrt{16+9} = \sqrt{25} = \mathbf{5} \text{ units}$$

$$BC = \sqrt{(1-9)^2+(4-10)^2} = \sqrt{(-8)^2+(-6)^2} = \sqrt{64+36} = \sqrt{100} = \mathbf{10} \text{ units}$$

(iii) Verification of Midpoint Theorem [1 Mark]
Comparing the two lengths:

$$\frac{1}{2}BC = \frac{1}{2} \times 10 = 5 \text{ units} = EF \implies \boxed{\mathbf{EF = \frac{1}{2}BC} \quad \text{(Hence Verified)}}$$

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